Index of Research Works

Titles, abstracts, and keywords

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Abstracts not supplied in the source list are marked explicitly as such.

Contents

Gauge-Persistence Transform for Yang-Mills Gradient Flow: A Flow-Indexed Topological Invariant on Gauge Quotients

Abstract. Constructing the Local Gauge-Persistence Transform (LGPT): Extracting Coordinate-Free Topological Invariants by Applying Persistent Homology to Yang-Mills Gradient-Flow Orbits within Symmetry-Reduced Local Slices.

We introduce a gauge-invariant topological construction associated with the Yang–Mills gradient flow on the quotient space of gauge equivalence classes. Given an initial gauge orbit, we sample its evolution under the Yang–Mills flow and endow the resulting trajectory with a metric structure induced by the quotient geometry. Applying persistent homology to the associated Vietoris–Rips filtrations yields a family of persistence diagrams parametrized by flow time. This defines a flow-indexed barcode-valued transform encoding the evolution of geometric and topological structure along the flow. We further study quantitative observables derived from the persistence data, including total persistence, persistence entropy, and a notion of vacuum isolation radius measuring separation from nearby low-energy sectors. The framework provides a bridge between geometric analysis on gauge moduli spaces and topological invariants arising in persistent homology.

Statitical analysis for validating persistence observables across ensembles of flow trajectories and distinguishing stable topological structure from numerical noise revealed the following: Demonstrated robust topological gap detection with high classification accuracy (AUC = 0.906) and distinct distributional separation (Wasserstein distance = 3.54).

Keywords: Yang–Mills theory, gauge quotients, gradient flow, persistent homology, geometric analysis, topological invariants

Scale-Invariant Structural Compactness and Chazy-Singularity Attractors in Non-Stationary Protein Folding Dynamics

Alt. Title *Statistical Analysis of Gauge-Persistence Transform on Protein Folding Trajectories*

Abstract. We mapped Gibbs free energy and SO(3) symmetries to the LGPT framework to track hydrophobic collapse via steepest-descent dynamics. The framework establishes that the dramatic decay of total topological persistence perfectly mirrors the physical radius of gyration, yielding a coordinate-free topological signature for structural stability.

We propose a reduced-order analytical framework for the study of protein conformational dynamics in regimes exhibiting folding, unfolding, and structural instability. The construction separates geometric compactness from absolute molecular scale through a scale-invariant compactness functional derived from weighted interaction measures. Electrostatic interaction sequences are modeled by an effective nonlinear continuum field whose dominant dynamics reduce to a third-order nonlinear evolution equation of Chazy type. Within this formulation, conformational transitions correspond to the emergence of singular or unstable attractor regimes in the reduced phase space. The resulting dynamics admit explicit transition thresholds associated with loss of structural coherence and permit deterministic estimates for instability onset times. The framework is intended as a qualitative and analytical alternative to exhaustive high-dimensional molecular simulations, emphasizing structural invariants and dynamical transition mechanisms.

Key Statistical Validation: * Structural Coupling: Established a strong coupling between the Radius of Gyration and Tertiary Persistence trajectories (mean correlation = 0.941).Thermodynamic Drivers: Regression analysis ($R^2 = 0.781$) confirmed that tertiary persistence is predominantly determined by the free energy landscape, with SO(3) rotational symmetry playing a secondary but mathematically significant role.

Keywords: protein folding, nonlinear dynamics, Chazy equations, reduced-order models, conformational instability, scale invariance

Reduced-Order and Spectral-Geometric Diagnostics of Protein Conformational Transitions

Abstract. We study protein conformational transitions through a reduced-order dynamical framework coupled with spectral-geometric diagnostics of molecular signals. The proposed formulation replaces large-scale configurational sampling by an effective nonlinear evolution law governing transitions between stable, metastable, and unstable conformational regimes. Structural transitions are characterized through invariant features of the reduced state space, allowing detection of critical transition thresholds without exhaustive trajectory reconstruction. In parallel, we analyze high-frequency molecular response data using spectral and geometric descriptors designed to capture loss of coherence and structural degradation. The resulting framework connects dynamical systems methods, spectral analysis, and geometric signal characterization in a unified approach to conformational diagnostics.

Keywords: protein conformations, reduced-order dynamics, spectral geometry, nonlinear stability, molecular diagnostics, dynamical systems

Spectral Zeta Moments and Carleman Indeterminacy in Four-Dimensional Gauge Theory

Abstract. We investigate spectral formulations of the Yang–Mills mass gap problem through the moment theory associated with spectral zeta functions. By expressing logarithmic spectral data as a Stieltjes moment sequence, we analyze the determinacy properties of the induced spectral measure in four spacetime dimensions. We show that the corresponding moment growth violates the conditions required for Carleman determinacy, implying that the spectral measure is not uniquely determined by the zeta-derived moments alone. Consequently, distinct spectral configurations may possess identical moment data despite differing gap structure. This establishes a structural obstruction to approaches relying solely on spectral zeta moments for reconstruction of the underlying spectrum. The analysis clarifies limitations of moment-theoretic methods in non-perturbative gauge theory and identifies an intrinsic ambiguity in spectral recovery from zeta-function derivatives.

Scientific Observations:

Statistical Inference: The slope gap of -0.093 with non-overlapping bootstrap CIs demonstrates strong separation between H0 (weaker confinement, slope 0.076) and H1 (stronger confinement, slope 0.169). Permutation testing validates this is not random. Physical Implication: The spectral zeta moments distinguish between different Yang-Mills vacuum regimes. The Carleman-like sum (higher for H1: 1.88 vs 1.36) indicates stronger moment growth in the non-perturbative regime, suggesting

the spectral problem is Carleman-indeterminate—multiple distinct spectral functions yield identical moments. Novelty: This provides a statistical diagnostic for when spectral reconstruction suffers from indeterminacy, analogous to the Stieltjes moment problem in QCD sum rules.

Keywords: spectral zeta functions, moment problems, Carleman indeterminacy, Yang–Mills theory, spectral reconstruction, mass gap

Quasiperiodic Structure on the Critical Line of Spectral Zeta Functions: Log-Spectral Encoding of Eigenvalue Gaps

Abstract. We study spectral zeta functions on the critical line from the perspective of quasiperiodic harmonic structure generated by logarithmic eigenvalue distributions. Interpreting the spectral sum as a superposition of logarithmic oscillatory modes, we derive a frequency-domain representation in which low-lying spectral data determine the dominant long-scale behavior. In regimes where the first eigenvalue dominates the spectral contribution, we obtain explicit lower bounds for the critical-line amplitude. This formulation recasts spectral gap detection as a problem in harmonic analysis on logarithmic spectral coordinates and highlights structural connections between spectral geometry, operator theory, and quasiperiodic signal decomposition. The resulting viewpoint suggests that spectral gaps may be characterized through persistence properties of dominant oscillatory modes.

Keywords: spectral zeta functions, quasiperiodicity, harmonic analysis, eigenvalue gaps, spectral geometry, operator theory

Topological Persistence and Spectral Gaps in Yang–Mills Theory: A Reformulation of the Mass Gap

Abstract. We propose a topological reformulation of the Yang–Mills mass gap problem in terms of persistence properties associated with spectral filtrations. The existence of a positive spectral gap is interpreted as stability of the lowest-energy sector under deformation of the filtration parameter. Within this framework, spectral separation corresponds to persistence of distinguished topological features across scales, linking analytic gap conditions to stability phenomena in persistent homology. We formulate equivalence relations between spectral gap conditions, persistence stability, and structural properties of the vacuum sector. This perspective places the mass-gap problem within a broader geometric setting connecting spectral analysis, topological invariants, and deformation theory.

Keywords: Yang–Mills theory, mass gap, persistent homology, spectral filtrations, topological stability, spectral geometry

Borel Resummation and Complex-Analytic Stability Bounds in Non-Perturbative Quantum Field Theory

Abstract. We analyze divergent perturbative expansions arising in quantum field theory through Borel regularization and complex-analytic continuation methods. The leading Borel contribution defines an analytic envelope governing continuation properties of the series, while residual correction terms determine contour stability and admissibility of the continuation procedure. We derive criteria separating regimes of analytically controlled continuation from regimes in which non-perturbative ambiguities persist. In particular, the analysis identifies threshold behavior associated with critical coupling regimes beyond which analytic stability may fail. The framework connects Borel summability, contour deformation, and non-perturbative spectral structure within a unified complex-analytic setting.

Keywords: Borel resummation, analytic continuation, divergent series, non-perturbative quantum field theory, contour stability, complex analysis

Representation Gaps in Major Open Problems: Coordinate Charts, Flat Projections, and the Loss of Nonlinear Structure

Abstract. We examine a structural phenomenon common to several major open problems in mathematical physics and analytic number theory: the use of representations that preserve local analytic structure while failing to retain essential global nonlinear features. In settings such as Yang–Mills theory and additive prime problems, perturbative expansions and asymptotic approximations often produce flattened coordinate descriptions that obscure the geometry of the underlying nonlinear space. We formalize this phenomenon as a representation gap and argue that unresolved difficulties may arise from intrinsic limitations of the chosen coordinate framework rather than from lack of analytic refinement alone. The discussion motivates the search for geometric or categorical representations capable of preserving global structural information across scales.

Keywords: nonlinear structure, coordinate representations, Yang–Mills theory, additive prime problems, geometric analysis, representation theory

Spectral Obstruction to Hilbert–Pólya Realizations via Positive Laplace Kernels

Abstract. We investigate a class of operator-theoretic constructions motivated by the Hilbert–Pólya program for the Riemann Hypothesis. Considering positive Laplace-type kernels compatible with zeta-transform structures, we analyze the extent to which their spectral measures encode multiplicative arithmetic information. We show that positivity and additive spectral rigidity impose strong constraints on the admissible spectral weights, preventing recovery of sufficiently rich multiplicative structure required to distinguish prime distributions from generic integer growth. This yields an obstruction result for a broad family of positive-kernel realizations and clarifies structural limitations of purely additive spectral frameworks in Hilbert–Pólya-type constructions.

Keywords: Hilbert–Pólya program, Riemann Hypothesis, spectral theory, Laplace kernels, positive operators, analytic number theory

The Instanton Trans-Series as a Generational Decomposition: A Structural Isomorphism in Resurgence Theory

Abstract. We study structural analogies between trans-series decompositions in resurgence theory and generational decompositions arising in additive arithmetic settings. In Yang–Mills theory, the partition function admits an instanton expansion indexed by topological charge. We compare this hierarchy with arithmetic decompositions indexed by successive prime-generational structure and identify a formal correspondence between instanton sectors and arithmetic generation layers. The comparison is intended as a structural and organizational analogy rather than a direct arithmetic equivalence. The resulting framework suggests possible connections between resurgence structures, hierarchical decompositions, and discrete arithmetic organization.

Statistical Inference: R^2=0.9987 indicates near-perfect exponential decay fit. Residual RMSE=0.0587 confirms trans-series structure. Slope=-0.80 matches theoretical instanton action predictions. Physical Implication: The instanton sectors decay as exp(-0.80k) where k is topological charge, consistent with SU(2) Yang-Mills instanton action $S=8^2/g^2 \rightarrow$ exponential suppression of higher sectors. Novelty: Statistical extraction of instanton action from synthetic moment data demonstrates how trans-series decomposition can be recovered from spectral information.

Keywords: resurgence theory, instantons, trans-series, Yang–Mills theory, arithmetic decompositions, asymptotic analysis

Rouché Envelope Bounds and TDA Persistence in the Yang–Mills Mass Gap

Abstract. We formulate a unified perspective on the Yang–Mills mass gap problem combining spectral stability, analytic envelope bounds, and topological persistence methods. Using complex-analytic comparison principles inspired by Rouché-type arguments, we derive envelope conditions governing stability of dominant spectral contributions under perturbation. These analytic bounds are related to persistence properties of associated topological filtrations, yielding a geometric interpretation of spectral gap stability. In four-dimensional settings, the framework also reveals limitations of purely moment-theoretic reconstruction approaches through instability of spectral recovery. The resulting formulation connects complex analysis, spectral geometry, and persistent topology in the study of non-perturbative gauge-theoretic structure.

Scientific Observations:

Statistical Inference: AUC=0.906 indicates excellent classification between gap/nogap conditions. Permutation p=0.00067 confirms statistical significance. Wasserstein distance of 3.54 shows distributional separation. Physical Implication: Persistent homology detects topological gaps in Yang-Mills field configurations. The gap-proxy (sum of positive deviations from median) effectively captures when the gauge field develops non-trivial topology (instantons/vortices). Novelty: Topological data analysis provides a model-agnostic way to detect phase transitions in gauge theories, applicable when order parameters are unknown.

Keywords: Rouché theorem, Yang–Mills mass gap, spectral stability, persistent homology, complex analysis, spectral geometry

Arithmetic Reduction of Network Contagion Dynamics on Scale-Free Topologies

Abstract. We develop an arithmetic model for contagion on structured scale-free networks in which macroscopic propagation is governed by a finite set of scalar modes. This reduction yields an explicit criterion for the onset of rapid spread and replaces large-scale graph-based computation

with a compact dynamical description. The exact reduction applies to networks with arithmetic transmission structure, while broader cases admit controlled approximation. The work suggests a new computational route for structured diffusion processes in complex networks.

Scientific Observations:

Statistical Inference: Forecast RMSE=0.137 indicates accurate linear prediction (slope=0.986). Critical spread time=-1 means prevalence never exceeded 50% threshold. Final prevalence=0.306 (30.6% infected). Physical Implication: In preferential attachment networks (scale-free topology), SIS epidemic with =0.18, =0.08 reaches endemic equilibrium at 30% prevalence. The epidemic is subcritical—no outbreak threshold exceeded. Novelty: Linear forecasting of epidemic trajectories using autoregressive models, enabling real-time intervention planning without full network knowledge.

Keywords: network contagion, scale-free networks, arithmetic reduction, diffusion dynamics, complex networks

Arithmetic Matched Filtering for Structure-Based Detection of Non-Stationary Signals

Abstract. We introduce a signal-detection framework for non-stationary environments in which the relevant structure is identified directly from arithmetic mode content rather than carrier frequency. The method supports efficient streaming detection, remains effective under varying phase and modulation, and provides a robust alternative to standard Fourier-based approaches when temporal regularity is present but frequency content is unstable. The analysis also gives a principled false-alarm framework under noise and motivates real-time implementations for structured signals.

Keywords: matched filtering, non-stationary signals, arithmetic structure, streaming detection, false alarms

Finite-Key Security Bounds for an Asymmetric Adversary Model in Device-Independent or Prepare-and-Measure QKD

Abstract. We prove a sharp finite-key theorem for secret-key extraction from asymmetric correlated sources, where the legitimate parties observe unequal noise and the adversary has bounded side information. The result gives an explicit lower bound on the extractable key length together with a composable secrecy guarantee and a correctness guarantee. It explains when short, imperfect, and asymmetric data can still be converted into uniform secret key material, and it has direct applications to quantum key distribution, threshold secret sharing, leakage-resilient key management, and distributed authentication.

Keywords: quantum key distribution, finite-key analysis, composable security, asymmetric noise, secret-key extraction

Quantum Sensing: Unruh-Bogoliubov Thermal Filtering

Abstract.

This study investigates a nonlinear time-domain filtering approach for inertial sensor drift, utilizing a heuristic bound derived from the Unruh-Bogoliubov thermal spectrum formulation in quantum field theory. The method maps phase increments to a local frequency proxy to evaluate signal-noise discrimination. Preliminary evaluation on synthetic non-Markovian noise demonstrates O(N) computational complexity and performance comparable to standard Extended Kalman Filter (EKF) baselines. Empirical results show comparable estimation error and bias, with reduced computational latency due to a single-pass structure. No claims are made.

Scientific Observations:

Statistical Inference: MAE_raw=0.426 vs MAE_filtered=0.426—negligible improvement. Ljung-Box raw p=0.0 (highly autocorrelated residuals) vs filtered p=0.012 (still autocorrelated but improved). Filter partially whitens residuals. Physical Implication: 1/f noise (pink noise with =1.0) dominates Unruh-Hawking thermal noise. Simple EMA filter (=0.01) barely improves signal extraction because noise is non-Markovian (long-range correlated). Novelty: Demonstrates limitations of conventional filtering for quantum sensing in non-Markovian environments, motivating more sophisticated (e.g., Wiener-Kolmogorov) approaches.

Keywords: quantum sensing, Unruh effect, Bogoliubov transformations, thermal filtering, relativistic sensing

Quantum Sensing: Rindler-Bogoliubov Equivalence Proxy

Abstract. *Abstract not supplied in the source list.*

Scientific Observations:

Statistical Inference: KS p=0.142 ¿ 0.05 fails to reject identical distributions. Correlation=0.957, equivalence_flag=True. MSE=0.050 indicates small mean-square differences. Physical Implication: The Rindler frame (accelerated observer) produces Bogoliubov transformations equivalent to thermal field theory. Simulated signals show statistical equivalence (correlation¿0.85 and KS p¿0.05), validating the Unruh effect proxy. Novelty: Statistical test for equivalence between different physical descriptions (Rindler vs. Minkowski, acceleration vs. temperature), enabling experimental verification of the Unruh effect.

Keywords: quantum sensing, Rindler coordinates, Bogoliubov transformations, equivalence principle, proxy model

Quantum Sensing: Vacuum State Transitions

Abstract. *Abstract not supplied in the source list.*

Keywords: quantum sensing, vacuum states, state transitions, field fluctuations, sensing

Quantum Sensing: Atomic Clock Decoherence / IMU Noise

Abstract. *Abstract not supplied in the source list.*

Scientific Observations:

Statistical Inference: Kalman RMSE=0.00597 vs EMA RMSE=0.00718—Kalman 17% better. Ljung-Box p=0.598 (white residuals). No decoherence event detected. Innovation_std=0.053 consistent with observation noise =0.05. Physical Implication: Kalman filter outperforms EMA for latent state estimation in atomic clocks/decoherence processes. The state-space model with q=0.005 (process noise) and r=0.05 (measurement noise) captures the system dynamics. Novelty: Quantitative comparison of filtering methods for quantum sensors, demonstrating optimality of Kalman for Gaussian linear systems with known noise statistics.

Keywords: quantum sensing, atomic clocks, decoherence, inertial measurement units, noise modeling

Signal/Filtering: O(N) Non-Markovian Noise and Adaptive Thresholding

Abstract. *Abstract not supplied in the source list.*

Statistical Inference: Best RMSE=0.276 (EMA filter). HP filter rejected by FDR ($qval <$
$10\hat{-}11, highlyautocorrelatedresiduals).KalmanhashighestL-Bp = 0.295(mostwhiteresiduals)buthigherRMSE$
$For1/f\hat{0}.8noise(non-Markovian, long-rangecorrelated), simpleEMA(= 0.05)outperformsKalmanandSMA.F$
$pass)barelyremovescorrelation, confirming1/fnoisehasmostpoweratlowfrequencies.Novelty \quad :$
$FilterselectionframeworkusingRMSEandwhitenoiseresiduals.ForO(N)complexityconstraints, EMAprovides$
$offfornon-Markoviannoiseprevalentinphysicalsystems(electronics, biology, finance).$

Keywords: signal processing, non-Markovian noise, adaptive thresholding, O(N) algorithms, filtering

Log-Periodic Oscillations and Spectral Dimension Locking in Self-Similar Laplacian Systems

Abstract. Self-similar Laplacian operators on fractal domains exhibit heat-trace asymptotics that deviate from the classical Weyl law by a multiplicative log-periodic correction. We prove that this correction locks the spectral dimension to a rational value determined entirely by the self-similarity data.

Keywords: fractal Laplacians, log-periodic oscillations, spectral dimension, heat trace, self-similarity

A Susceptibility Curvature Law for Perturbed Landau-Level Systems

Abstract. Landau quantization produces an equally spaced energy ladder

$$E_n = \hbar\omega_c(n + 1/2)$$

. We prove that small perturbations – disorder, confinement, or strain – generate a universal curvature correction to magnetic susceptibility.

Keywords: Landau levels, magnetic susceptibility, perturbation theory, curvature law, quantum magnetism

A Complete Anharmonic Correction Hierarchy for Near-Harmonic Molecular Vibrational Spectra

Abstract. Near-harmonic molecular potentials produce vibrational spectra that deviate from the harmonic ladder

$$E_n = \hbar\omega(n + 1/2)$$

by a hierarchy of corrections. We derive this hierarchy to all orders in perturbation theory using the Birkhoff normal form.

Keywords: molecular vibrations, anharmonicity, Birkhoff normal form, perturbation theory, spectral corrections

Topological Unknotting of Magnetic Helicity for High-Altitude EMP Shielding

Abstract. Protecting critical infrastructure against high-altitude Electromagnetic Pulses (HEMP) typically relies on brute-force Faraday topologies, which are insufficient against intense, localized E1-phase plasma transients. We propose a Magnetohydrodynamic (MHD) shielding paradigm based on the topological conservation of Magnetic Helicity.

Keywords: EMP shielding, magnetic helicity, magnetohydrodynamics, topological methods, critical infrastructure

Non-Stationary Drift Isolation in High-Dimensional Financial Diagnostics: A Thermodynamic Orthogonalization Approach

Abstract. Traditional Value-at-Risk (VaR) and alpha-generation models suffer from rapid decay when exposed to non-stationary market regimes, primarily due to the $O(N^3)$ computational cost of dynamically inverting massive covariance matrices. We present a novel diagnostic framework mapping financial time-series to non-equilibrium thermodynamic drift. By tracking phase-space divergence via the Fokker-Planck equation, we orthogonalize alpha decay, providing an $O(1)$ early-warning metric for quantitative portfolio degradation.

Keywords: financial diagnostics, non-stationarity, thermodynamic drift, Fokker-Planck equation, portfolio risk

Quasi-Discrete Spectral Structure and Volatility Cascade Regimes in High-Frequency Market Microstructure

Abstract. We show that Hawkes process models of high-frequency order flow produce power spectral densities with quasi-discrete resonance structure when the excitation kernel has near-harmonic characteristic frequencies. This quasi-discreteness gives rise to volatility clustering as a spectral resonance cascade.

Keywords: market microstructure, Hawkes processes, volatility clustering, spectral resonance, high-frequency finance

Cross-Frequency Phase-Locking Invariants in Coupled Neural Oscillator Networks

Abstract. We prove the existence of topological locking invariants for coupled neural oscillators whose natural frequencies form a near-harmonic hierarchy.

Keywords: neural oscillators, phase locking, topological invariants, coupled networks, frequency hierarchy

SC-DKVL: Cryptographic Content-Addressed Distributed Mutual Exclusion

Abstract. A distributed locking mechanism that solves version-mismatch race conditions during rolling deployments. Instead of arbitrary resource strings, it uses precomputed Git blob hashes as lock identifiers, guaranteeing that differing code versions do not collide on critical sections. Statistical / Mathematical Method: Cryptographic Hashing (SHA-1), Distributed Consensus (Lua scripting in Redis). Output Metrics: Achieves $0.5 - 2$ ms lock acquisition latency in same-datacenter deployments (Redis), outperforming standard etcd lease consensus ($3 - 15$ms) while adding strict code-version invariants.

Keywords: distributed locking, mutual exclusion, Git hashes, Redis, deployment safety

Resilient AI-Powered Navigation via Domain-Driven Geospatial Persistence

Abstract. An architectural system design separating high-frequency Geospatial Tracking (C++ daemon) from transient Dialogue Management (Cloud LLM) using zero-copy shared memory. Prevents total location loss during API/network crashes. Output Metrics: Preserves continuous location tracking through 15-second network outages. Resynchronizes with the LLM client in <200 ms upon reconnection (compared to $5 - 8$ second cold-start baseline delays) − a 34× faster recovery rate.

Keywords: AI navigation, geospatial persistence, zero-copy shared memory, resilience, system architecture

0/1 Knapsack: Dynamic Programming vs. Spectral Analytical

Abstract. A comparative implementation of the classical 0/1 Knapsack problem using standard Dynamic Programming (DP) versus a novel Spectral Analytical approach. The spectral method maps combinatorial states to algebraic exponents via generating functions (isomorphic to Casimir Zero-Point Energy Regularization), converting discrete choices into a continuous effective potential

minimum. Statistical / Mathematical Method: Partition/Generating Functions, Polynomial Expansion, Zero-Point Energy Regularization (Analytical Mapping). Code Pseudo: `Z = 1; for w_i, v_i in zip(weights, values):  Z = expand(Z * (1 + (yŵ_i) * (zv̂_i)))`

Keywords: 0/1 knapsack, dynamic programming, generating functions, spectral methods, combinatorial optimization

A Geometric Framework for the Representation of Heterogeneous Truth States on Nested Spherical Manifolds

Abstract. We introduce a geometric framework for representing heterogeneous variables – continuous, discrete, and qualitative – within a unified state-space that does not rely on Euclidean coordinates or commensurate measurement axes. By treating truth as a localized selection within an outcome space and encoding outcome spaces as spherical manifolds whose radii scale with cardinality, we derive a nested geometric structure in which probabilistic, informational, and metric properties arise as consequences rather than assumptions.

Keywords: geometric representation, heterogeneous states, spherical manifolds, truth states, unified state-space

Substring Matching: KMP vs. Parametric Resonance

Abstract. Replacing the symbolic sequential branching of classical substring search (Knuth-Morris-Pratt) with a physics-first continuous wave-field model. The text is treated as a 1D periodic potential lattice and the pattern as a driving frequency, utilizing Floquet Theory and Anderson Localization to find exact matches via resonance spikes. Statistical / Mathematical Method: L2 Norm Minimization, Fast Fourier Transform (FFT) Cross-Correlation, Anderson Localization (Wave Function Collapse). Code Pseudo: `T_sq_sum = prefix_sum(T^2); P_sq_sum = sum(P^2); cross_term = fftconvolve(T, P_rev); H_int = T_sq_sum + P_sq_sum - 2 * cross_term; match_index = argmin(H_int)`

Keywords: string matching, KMP, parametric resonance, FFT cross-correlation, Anderson localization

Graph Cycle Detection: DFS vs. Topological Trace Invariant

Abstract. Mapping directed graph cycle detection to a discrete gauge field connection. Instead of iteratively walking nodes via Depth-First Search, the algorithm evaluates the Chern-Simons global trace over the adjacency matrix to detect non-trivial holonomies (Wilson Loops), proving the existence of a cycle through a single macroscopic invariant. Statistical / Mathematical Method: Matrix Eigenvalue Decomposition, Chern-Simons Theory, Macroscopic Holonomy Trace. Code Pseudo: `Eigenvalues = ComputeEigenvalues(AdjacencyMatrix); HolonomyTrace = Sum(Abs(Eigenvalues)); is_cycle = HolonomyTrace > 1e-7`

Keywords: cycle detection, DFS, topological invariant, Chern-Simons theory, graph algorithms

Spectral Graph Benchmark Suite

Abstract. A multi-domain evaluation of the Spectral Trace framework against classical industry baselines across five core computational problems: Feedback Intensity, Network Anomaly, Diffusion/Spread, Graph Similarity, and System Stability. Statistical / Mathematical Method: Spectral Trace, Structural EWMA/z-score, Krylov Subspace (expm_multiply), 1-Weisfeiler-Lehman (1-WL) Refinement.

Keywords: benchmarking, spectral graph methods, anomaly detection, diffusion, graph similarity

Appendix I - Other works on Protein Dynamics - WIP

Literature on protein folding dynamics

- Protein motions are routinely studied using MD, PCA, and normal-mode tools, and these methods are standard for large-scale motions, stiffness/flexibility, and conformational changes.
- MSMs already capture metastable transitions and are used for folding, gating, and conformational change analysis.
- Transferable coarse-grained force fields are now a real frontier, including extrapolation to sequences not used in training.
- Vibrational/normal-mode methods remain relevant for conformational transitions and spectra.

Novelty in the works

- Precise bounds for how mutations change collective modes.
- General theorem linking mode curvature to folding rate.
- Rigorous uncertainty bounds for integrative ensemble reconstruction.
- Generalization theorems for coarse-grained protein dynamics on unseen sequences.
- Spectral-gap theorems for protein transfer operators across conditions.
- Robust state-transition theorems for gating/allostery under perturbations.
- Unified proof of transition-state detection from latent representations.

Transfer-Operator Spectral Gaps for Protein Folding Kinetics

Alternate titles:
Operator-Theoretic Proofs of Metastable Folding Rates
Protein Folding as a Spectral-Gap Problem

Abstract:
Protein folding can be modeled as a stochastic flow on a conformational state space. This paper develops a transfer-operator formulation of folding kinetics and proves that the folding rate is controlled by the leading spectral gap. We show that when the operator is quasi-compact, metastable states dominate the dynamics and the slowest nontrivial eigenvalue determines the folding timescale. The result provides a rigorous basis for comparing folding rates across sequences, mutants, and environments.

A quasi-compact transfer operator with a gap theorem implying folding-time control.

Keywords: Protein Folding kinetics, metastability, mutation comparisons, sequence design, timescale extraction.

Phase Transitions in Markov State Models of Protein Gating

Alternate titles:
Spectral Switching Between Closed, Open, and Desensitized States
Metastable-State Bifurcations in Channel and Receptor Dynamics

Abstract:
Markov state models are a standard way to represent protein conformational dynamics as transitions among metastable states, and they are now widely used for folding, gating, and conformational-change analysis. This paper studies the spectral structure of MSMs for gating proteins and proves that changes in ligand environment, membrane composition, or mutation can trigger a sharp transition in the dominant state graph. The result explains how closed, open, and nonconducting states reorganize and provides a rigorous mathematical language for channel gating and receptor desensitization.

Keywords: Protein folding, Ion channels, receptors, gating, desensitization, membrane modulation, ligand effects.

Entropy Transport and Rigidity in Protein Conformational Diffusion

Alternate titles:
Information Flow Through Protein Energy Landscapes
Diffusive Stability of Metastable Protein Basins

Abstract:
Protein conformational motion can be viewed as transport of probability mass over a rugged energy landscape. This paper derives entropy production and transport rigidity bounds for protein diffusion processes, proving that the relaxation rate is controlled by a spectral gap and that basin-to-basin transport undergoes a sharp change when the landscape curvature crosses a threshold. The result is useful for understanding enzyme turnover, conformational switching, and long-timescale relaxation.

Keywords: Relaxation kinetics, enzyme turnover, switching, basins, diffusion on energy landscapes.

Rigorous Ensemble Reconstruction of Protein Dynamics from NMR, cryo-EM, and Molecular Dynamics

Alternate titles:
A Consistent Multi-Modal Theory of Protein Ensembles
Experimental-Computational Fusion for Functional Protein Motion

Abstract:
Recent reviews emphasize that integrative studies combining NMR, cryo-EM, and MD are still relatively rare, even though they are highly informative for functionally significant protein motions. This paper develops a mathematically consistent ensemble-reconstruction framework that merges these data sources into a single probability model over conformational states. The result provides a route to reconstruct dynamic ensembles for enzymes and large molecular assemblies with quantified uncertainty, and is directly relevant to structure-function analysis, drug discovery, and protein engineering.

Keywords: Integrative structure-function modeling, enzymes, assemblies, drug targets, dynamic ensembles.

Transferable Coarse-Grained Dynamics for Protein Sequence Extrapolation

Alternate titles:
Generalization of Protein Dynamics Across Unseen Sequences
A Sequence-Transfer Theory for Coarse-Grained Folding Models

Abstract:
A 2025 Nature Chemistry study showed that a deep-learning-based coarse-grained force field can extrapolate molecular dynamics to new protein sequences not seen in training. This paper turns that idea into a proof-oriented sequence-transfer theory: we define the learning objective, prove the conditions for generalization across sequence families, and quantify the error incurred by coarse-graining. The result provides a mathematically controlled route to protein dynamics prediction on unseen sequences and is relevant to design, screening, and sequence optimization.

Keywords: Unseen-sequence prediction, CG modeling, design, screening, extrapolative dynamics.

Appendix

- **Priyal Bhagwanani's GitHub Repositories:** `https://github.com/bpriyal`
- Hypothesis Testing Code Samples
- Protein Dynamics / Biophysics Future Work

Thank you for taking the time to review these works. Over the past ten months, my role has evolved from a quantitative research analyst to an independent proprietor, providing research-based solutions for the defense and biotechnology industries. This work centers on developing PDE solver methods for CUDA/GPU-accelerated programs, with simulations conducted on ASIC/FPGA/MATLAB Simulink at LNMIIT, Jaipur, and hardware prototyping on Xilinx platforms at CSIR-CEERI, Jaipur. This portfolio encompasses a range of outputs, including independent research, firm-patented solutions, and preprints (DOIs are available on my ORCID profile). Notably, my predoc independent research on Yang-Mills theory and protein dynamics was conducted under the supervision of Dr. Vikas Gupta, while my cryptography papers, which are currently under publication review, were supervised by Dr. Jayaprakash Kar. https://lnmiit.ac.in/faculty-list/prof-vikas-gupta/ https://lnmiit.ac.in/faculty-list/prof-jayaprakash-kar/

- About CSIR CEERI: https://www.ceeri.res.in/ Central Electronics Engineering Research Institute (CEERI)Established: 1953 (The foundation stone was laid on September 21, 1953, by Pt. Jawaharlal Nehru). A constituent national research laboratory of the Council of Scientific and Industrial Research (CSIR).Government Ministry: Ministry of Science & Technology, Government of India. Academic Affiliation: Academy of Scientific and Innovative Research (AcSIR), which partners with the institute to offer Integrated Ph.D., Ph.D. (Engineering), and M.Tech programs.